

SDS 8102: Advanced Mathematical Statistics

Homework 2

Fall 2026

MBZUAI

Due Date: Sep 25, 2026 (11:59 PM)

Guidelines:

- Start working on the homework early. If you have questions regarding notation, problem statements, or possible errors, please contact us as soon as possible. You *must begin your email* with the subject line HW [homework number]-CLARIFICATION-SDS 8102 (FALL 2026). Without the subject line, the email may get ignored.
- Prepare a $\text{\LaTeX}$ file for each assignment and only submit your final PDF file in the LMS system. Make sure *your name is clearly written* in the header. Start the solution to each main problem on a *new page*. Any code used in your solutions should be included in an appendix. Submit one combined PDF containing all of your solutions.
- The problems may be adapted from course materials, textbooks, and other standard references. If an external resource plays a substantial role in developing your solution, *please cite it appropriately*.
- The goal of the problems is to develop both intuition and technical skills. You may explore beyond what is explicitly asked. If you introduce additional assumptions, clearly state them. You are also welcome to include short remarks about interesting observations that arise while solving the problems.
- Problems or parts marked [**Bonus**] are optional and are not required for full credit. If you attempt a bonus problem, clearly label the corresponding solution.
- Some problems contain hints. These are intended only as suggestions for getting started; you are free to use a different approach.
- Solutions will be evaluated primarily on mathematical correctness and clarity of reasoning. There is no need to spend excessive effort on minor presentation details.
- **AI use:** AI-generated solutions are strictly discouraged in this course. You may use AI for minor polishing of the presentation. However, you should produce the core logic by yourself.

1 Unbiasedness, Completeness, and Admissibility in the Gaussian Model (10 points)

Let

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, 1), \quad \mu \in \mathbb{R},$$

and suppose that the parameter of interest is

$$\tau(\mu) = |\mu|.$$

This problem illustrates that the existence of a complete sufficient statistic does not guarantee the existence of an unbiased estimator of every function of the parameter. It also highlights the distinction between unbiasedness and admissibility.

1. **A complete sufficient admissible statistic.** [5 points]

Show that

$$T = \overline{X}$$

is sufficient and complete for μ . Also, show that it is admissible.

2. **Analyticity of Gaussian expectations.** [5 points]

Let

$$X \sim \mathcal{N}(\mu, 1),$$

and suppose that

$$\mathbb{E}_\mu[|g(X)|] < \infty \quad \text{for every } \mu \in \mathbb{R}.$$

Define

$$m(\mu) = \mathbb{E}_\mu[g(X)].$$

Using the likelihood-ratio representation

$$m(\mu) = \exp\left\{-\frac{\mu^2}{2}\right\} \mathbb{E}_0[g(X)e^{\mu X}],$$

formally expand

$$e^{\mu X} = \sum_{k=0}^{\infty} \frac{(\mu X)^k}{k!}.$$

Show that, for every $r > 0$,

$$\mathbb{E}_0[|g(X)|e^{r|X|}] < \infty.$$

Hint: Use

$$e^{r|X|} \leq e^{rX} + e^{-rX},$$

together with the identities

$$\mathbb{E}_0[|g(X)|e^{rX}] = e^{r^2/2} \mathbb{E}_r[|g(X)|]$$

and

$$\mathbb{E}_0 [|g(X)|e^{-rX}] = e^{r^2/2} \mathbb{E}_{-r} [|g(X)|].$$

Deduce that the interchange of expectation and summation is justified and that

$$m(\mu) = \exp \left\{ -\frac{\mu^2}{2} \right\} \sum_{k=0}^{\infty} \frac{\mu^k}{k!} \mathbb{E}_0 [g(X)X^k].$$

Show that the power series above has infinite radius of convergence. Conclude that m is differentiable on $\mathbb{R}$.

Finally, deduce that there cannot exist an unbiased estimator of $|\mu|$, because the function

$$\mu \mapsto |\mu|$$

is not differentiable at $\mu = 0$.

3. Admissibility without unbiasedness.

[Bonus]

Consider estimating $|\mu|$ under squared-error loss

$$L(\mu, \delta) = (\delta - |\mu|)^2.$$

Consider the constant estimator

$$\delta_0(X) = 0.$$

(a) Show that

$$R(0, \delta_0) = 0.$$

(b) Suppose that an estimator δ dominates δ_0 . Show that $R(0, \delta) = 0$, and hence

$$\delta(X) = 0 \quad P_0\text{-almost surely.}$$

(c) The Gaussian distributions P_μ and P_0 are mutually absolutely continuous. Use this fact to show that

$$\delta(X) = 0 \quad P_\mu\text{-almost surely}$$

for every μ . Conclude that δ_0 is admissible.

Briefly explain why this result demonstrates that admissibility and unbiasedness are fundamentally different properties.

2 Two-Sided Testing and Confidence Intervals in the Gaussian Model (10 points)

Let

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2),$$

where $\sigma^2 > 0$ is known. Consider testing

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0.$$

1. **The likelihood-ratio test.** [4 points]

Derive the likelihood-ratio test and show that its rejection region can be written as

$$\left\{ \frac{\sqrt{n} |\bar{X} - \mu_0|}{\sigma} > z_{1-\alpha/2} \right\},$$

where z_q denotes the q -quantile of the standard normal distribution.

Show that this test has size α .

2. **Inverting the test.** [3 points]

For every $\mu_0 \in \mathbb{R}$, let $A(\mu_0)$ denote the acceptance region of the level- α likelihood-ratio test above. Define

$$C(X) = \{\mu_0 \in \mathbb{R} : X \in A(\mu_0)\}.$$

Show that

$$C(X) = \left[\bar{X} - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right].$$

Verify directly that

$$\Pr_{\mu} \{\mu \in C(X)\} = 1 - \alpha \quad \text{for every } \mu \in \mathbb{R}.$$

3. **Why is the test not UMP?** [3 points]

Fix $\mu_1 > \mu_0$. Use the Neyman–Pearson lemma to show that the most powerful level- α test of

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu = \mu_1$$

rejects for sufficiently large values of $\bar{X}$.

Similarly, show that if $\mu_1 < \mu_0$, the most powerful test rejects for sufficiently small values of $\bar{X}$.

Explain why no single level- α test can be most powerful against both types of alternatives. Conclude that there is no UMP level- α test for

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0.$$

3 Bayes Estimation and the Impossibility of an Unbiased Bayes Rule (10 points)

Let

$$X | \theta \sim \mathcal{N}(\theta, 1), \quad \theta \sim \mathcal{N}(0, \tau^2),$$

where $0 < \tau^2 < \infty$. Suppose that θ is estimated under squared-error loss

$$L(\theta, a) = (a - \theta)^2.$$

1. **The Bayes estimator.**

[3 points]

Find the posterior distribution of θ given X , and show that the Bayes estimator is

$$\delta_\tau(X) = \frac{\tau^2}{1 + \tau^2} X.$$

Compute its Bayes risk.

2. **Frequentist bias.**

[2 points]

Treating θ as fixed, compute

$$\mathbb{E}_\theta[\delta_\tau(X)].$$

Show that δ_τ is biased for every finite τ^2 , except at $\theta = 0$.

What happens to $\delta_\tau(X)$ as $\tau^2 \rightarrow \infty$? Explain why this limiting argument does not produce a Bayes estimator under a proper prior.

3. **Can a proper Bayes estimator also be unbiased?**

[5 points]

Consider a general statistical model in which

$$\Theta \sim \pi, \quad X \mid \Theta = \theta \sim P_\theta,$$

where π is a proper prior with

$$\mathbb{E}_\pi[\Theta^2] < \infty.$$

Under squared-error loss, the Bayes estimator is

$$\delta_\pi(X) = \mathbb{E}[\Theta \mid X].$$

Suppose, in addition, that δ_π is unbiased:

$$\mathbb{E}_\theta[\delta_\pi(X)] = \theta \quad \text{for } \pi\text{-almost every } \theta.$$

(a) Use the defining property of conditional expectation to show that

$$\mathbb{E}[(\delta_\pi(X) - \Theta)\delta_\pi(X)] = 0.$$

(b) Use the assumed unbiasedness to show that

$$\mathbb{E}[(\delta_\pi(X) - \Theta)\Theta] = 0.$$

(c) Deduce that

$$\mathbb{E}[(\delta_\pi(X) - \Theta)^2] = 0.$$

Conclude that

$$\delta_\pi(X) = \Theta \quad \text{almost surely.}$$

(d) Explain why this is impossible in the Gaussian model above when $0 < \tau^2 < \infty$. More generally, explain why an estimator cannot be both unbiased and Bayes under a proper prior unless the data determine Θ exactly.

4 Bayesian Interpretations of Ridge Regression and the Lasso (12 points)

Consider the Gaussian linear model

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}_n(0, \sigma^2 I_n),$$

where $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$, and $\sigma^2 > 0$ is known.

1. Ridge regression from a Gaussian prior.

[4 points]

Suppose that the coordinates of β are independent with

$$\beta_j \sim \mathcal{N}(0, \tau^2), \quad j = 1, \dots, p.$$

(a) Show that the negative log-posterior density, up to terms that do not depend on β , is

$$\frac{1}{2\sigma^2} \|Y - X\beta\|_2^2 + \frac{1}{2\tau^2} \|\beta\|_2^2.$$

(b) Deduce that the posterior mode is the ridge estimator

$$\hat{\beta}_{\text{ridge}} = \arg \min_{b \in \mathbb{R}^p} \left\{ \frac{1}{2} \|Y - Xb\|_2^2 + \frac{\lambda}{2} \|b\|_2^2 \right\}, \quad \lambda = \frac{\sigma^2}{\tau^2}.$$

(c) Find the posterior distribution of $\beta \mid Y$, and show that

$$\mathbb{E}[\beta \mid Y] = (X^\top X + \lambda I_p)^{-1} X^\top Y.$$

Explain why, in this case, the posterior mean and posterior mode are the same.

2. The lasso from a Laplace prior.

[4 points]

Suppose instead that the coordinates of β are independent with Laplace densities

$$\pi(\beta_j) = \frac{1}{2b} \exp \left\{ -\frac{|\beta_j|}{b} \right\}, \quad b > 0.$$

(a) Show that the negative log-posterior density, up to terms not depending on β , is

$$\frac{1}{2\sigma^2} \|Y - X\beta\|_2^2 + \frac{1}{b} \|\beta\|_1.$$

(b) Deduce that the posterior mode is the lasso estimator

$$\hat{\beta}_{\text{lasso}} = \arg \min_{b \in \mathbb{R}^p} \left\{ \frac{1}{2} \|Y - Xb\|_2^2 + \lambda \|b\|_1 \right\}, \quad \lambda = \frac{\sigma^2}{b}.$$

3. **Shrinkage under an orthogonal design.**

[4 points]

Assume that

$$X^\top X = I_p$$

and define

$$Z = X^\top Y.$$

Show that the ridge estimator satisfies

$$\hat{\beta}_{\text{ridge},j} = \frac{Z_j}{1 + \lambda},$$

whereas the lasso estimator satisfies

$$\hat{\beta}_{\text{lasso},j} = \text{sign}(Z_j)(|Z_j| - \lambda)_+.$$

Compare the two forms of shrinkage. In particular, explain why ridge regression generally does not set coefficients exactly equal to zero, whereas the lasso can produce sparse estimates.

4. **Posterior mean versus posterior mode.**

[Bonus]

Under squared-error loss

$$L(\beta, a) = \|\beta - a\|_2^2,$$

the Bayes estimator is the posterior mean.

- (a) Explain why ridge regression is a Bayes estimator under squared-error loss.
- (b) Is the lasso generally the Bayes estimator under squared-error loss for the Laplace prior? Explain.
- (c) The Laplace prior has a continuous density and assigns probability zero to the event $\{\beta_j = 0\}$. Nevertheless, the lasso posterior mode may satisfy

$$\hat{\beta}_{\text{lasso},j} = 0.$$

Explain why there is no contradiction.